MIL-M-45094C(Ord)
28 February 1962

SUPERSEDING
MIL-M-45094B(Ord)
2 June 1960

MILITARY SPECIFICATION

MOUNT, TELESCOPE: M107

1. SCOPE

1.1 This specification covers one type of instrument known as Mount, Telescope: M107, hereinafter referred to as "mount".

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

NN-P-515	Plywood, Container Grade
UU-T-106	Tape, Masking; Paper (Pressure-Sensitive)
UU-T-111	Tape; Paper Gummed (Sealing and Securing)
PPP-B-601	Boxes; Wood, Cleated-Plywood
PPP-B-621	Boxes; Wood, Nailed and Lock-Corners
PPP-P-291	Paperboard; Wrapping, Cushioning
PPP-T-97	Tape; Pressure-Sensitive Adhesive, Filament Reinforced

Military

MIL-P-116	Preservation; Methods of
MIL-B-121	Barrier Material; Greaseproofed, Flexible (Waterproofed)
MIL-P-130	Paper, Wrapping, Laminated and creped
MIL-B-131	Barrier Material; Water Vaporproof, Flexible

FSC 1240

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Military

MIL-L-3150	Lubricating-oil; Preservative, Medium
MIL-A-5092	Adhesive; Rubber (Synthetic and Reclaimed Rubber Base)
MIL-R-6855	Rubber; Synthetic, Sheet, Molded, and Extruded, for Aircraft Applications
MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of
MIL-I-45208	Inspection requirements, General Specification for
MIL-I-45607	Inspection Equipment, Supply and Maintenance for Ordnance

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property

DRAWINGS

Ordnance Corps

F8603300	Mount, Telescope: M107
LSQAP-8603300	List of Supplementary Quality Assurance Provisions for Mount Telescope: M107
IEL-8603300	Inspection Equipment List for Mount Telescope: M107

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement function should be obtained from the procuring activity or as directed by the contracting officer).

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3. REQUIREMENTS

3.1 Production model.- At the beginning of production, one mount representative of production, shall be made available to the Government to determine if the mount meets all requirements of this specification.

3.2 Materials.- Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.3 Construction.- The mount shall be manufactured in accordance with Drawing F8603300, "Mount, Telescope" M107" and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical characteristics, performance and permanence to insure inherent quality and performance capabilities prior to final acceptance by the Government.

3.4 General specifications.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specification MIL-F-13926:

(a) Order of precedence
(b) Protective finished surfaces
(c) Special accuracy
(d) Finish, painting and protective
(e) Workmanship

3.5 Environmental.

3.5.1 Operating temperatures.- The mount shall meet the applicable requirements of 3.9 while exposed to and thermally stabilized at ambient temperatures of plus 150 degrees and minus 40 degrees Fahrenheit.

3.5.2 Vibration.- The mount shall be vibrated at a constant frequency of 30 cycles per second with an amplitude of 1/16 inch (1/8 inch total excursion) for a period of 5 minutes plus or minus 15 seconds. Subsequent to vibration there shall be no loose or missing parts and the mount shall meet all requirements of this specification.

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3.6 Alinement.- With the axis through the telescope mounting bearings forming an angle of 45 degrees, within 2 minutes to the front mounting surface and parallel to the vertical mounting surface within 2 minutes, the mount shall meet the following requirements.

3.6.1 Deflection movement.- The deflection boresight knob shall provide a minimum of 20 mils movement on each side of the position established in 3.6.

3.6.2 Elevation movement.- The elevation boresight knob shall provide a minimum of 20 mils in depression and 20 mils in elevation.

3.7 Backlash.- Backlash in the elevation and deflection boresight movement shall not exceed 0.1 mil.

3.8 Boresight locks.- With the boresight locks secured in the locked position with 60 pound inches of torque, there shall be no movement in deflection or elevation when a torque of 25 pound inches is applied to the boresight knobs. Locking and unlocking of the boresight locks with 60 pound inches torque shall not change the deflection and elevation alinement of the mount.

3.8.1 Elevation lock.- When the elevation locking lever is rotated counterclockwise from the locked position to the 1 o'clock position there shall be free elevation movement when the elevation boresight knob is rotated.

3.9 Operating knobs.

3.9.1 Torque.- The running torque required to rotate the elevation and deflection boresight knobs shall not exceed 12 pound inches at the standard ambient temperature (plus 60 degrees to 90 degrees Fahrenheit) and shall not exceed the value obtained at standard ambient temperature by more than 6 pound inches when subjected to extreme temperatures of 3.5.1.

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3.9.2 Force.- The force required to disengage the elevation and deflection boresight knobs shall not exceed 8 pounds at standard ambient temperatures, and shall not exceed the value obtained at standard ambient temperatures by more than 4 pounds when subjected to extreme temperatures of 3.5.1.

3.10 Identification marking.- Identification marking, unless otherwise specified, shall be in accordance with Standard MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection shall be in accordance with Specification MIL-I-45208.

4.2 Production model.- Unless otherwise specified in the contract, one mount shall be selected by the Government inspector as soon as practicable after the start of production and shall be inspected by the contractor under the surveillance of the Government. This inspection shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract. The model shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

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4.3 Lot formation.- Inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.3.1 Defective units and lots shall be processed as specified in Standard MIL-STD-105.

4.4 Inspection of components.- Inspection requirements for component parts and subassemblies are contained herein and on Supplementary Quality Assurance Provisions (SQAP) listed in Supplementary Quality Assurance Provisions List (LSQAP)-8603300.

4.4.1 Contents of SQAP.- The contents of SQAP shall represent the minimum inspection requirements and shall be the basis for acceptance or rejection by lot of those components covered by a SQAP. The Government reserves the right, however, to inspect for the requirements of all drawings and specifications and to reject individual non-conforming items.

4.4.1.1 Contractor's inspection for all components not covered by SQAP.- Inspection for these components shall be in accordance with the inspection procedures of the written inspection system required by Specification MIL-I-45208. The system shall include procedures for checking all positional and functional characteristics shown on the drawings which indicate precision requirements and have a bearing on the functional accuracy and performance of the mount.

4.5 Material.- Control of material shall be in accordance with test data requirements of Specification MIL-I-45208.

4.6 Inspection equipment.- Inspection equipment shall conform to drawings listed in IEL 8603300, except as provided for under the alternative inspection procedures and inspection equipment requirement of Specification MIL-I-45208. Provisions of MIL-I-45607 shall apply to test equipment provided by the Government or provided by the contractor and utilized in the Government approved inspection system. The Government reserves the right to use all approved test equipment both Government furnished and contractor acquired for verification inspection.

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4.6.1 Accuracy requirements.- Inspection equipment which incorporates features for measuring values or sizes established by product requirements shall be capable of measuring those values within an accuracy of ten percent of the specified tolerance except as otherwise specified in 4.6.2. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.6.2 Inspection equipment design requirements.

4.6.2.1 Device to assure alinement and to test deflection movement, elevation movement backlash, boresight locks.- The device shall permit positioning the mount in a horizontal position as shown on Drawing F8603300 so that the front mounting surface is at 45 degrees plus or minus 30 seconds and the rear mounting surface is vertical within 30 seconds. The mount shall be secured to the device by means of the mounting bolt holes provided on the mount. The provisions incorporated on the vertical mounting surface of the device shall permit positioning the mount by means of the .3740 inch diameter hole located 10.286 inch less 1.266 inch from the axis of the 7/16 diame er hole in the front mounting surface shown on Drawing F8603269, Sheet 1, and located 2.665 inch plus 1.344 inch down from the axis of the same 7/16 diameter drill hole. Each mounting surface shall have a machine finish of 32 micro inch root mean square or better and each shall have a surface flatness of .0005 over their entire surfaces.

4.6.2.2 Sighting device.- With the mount thus positioned in the device, a sighting de ice utilized to test deflection movement, elevation movement, backlash and boresight locks shall be centrally located within the telescope mounting bearings positioned by means of the positioning keyway and secured by means of the clamping device provided on the mount. No shake, shift, or other movement which might contribute to an unrealistic test shall be permitted on the devices.

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4.6.2.3 Test equipment used to test boresight knob and boresight lock torque requirements.- The device shall consist of a tester and appropriate adapters. The device shall have a minimum range of from zero to seventy pound inches the accuracy of which shall be plus or minus 1/4 pound inches throughout the entire zone of excursion in both the clockwise and counterclockwise direction.

4.6.2.4 Test equipment used to test boresight knob force requirements (pull).- The device shall consist of a tester and appropriate adapters. The device shall have a minimum range of from zero to ten pounds the accuracy of which shall be plus or minus 1/4 pound.

4.6.2.5 Test equipment used to test vibration requirements.- This equipment shall conform to the accuracy requirements specified in Specification MIL-F-13926.

4.6.2.6 Test equipment used to test temperature requirements.- This equipment shall meet the accuracy requirement specified in Specification MIL-F-13926.

4.7 Tests.

4.7.1 Classification of defects.- The classification of defects in Tables I, II, III and IV shall constitute the minimum inspection requirements to be performed prior to acceptance or rejection by lots. The Government reserves the right to inspect for any or all applicable requirements and to reject individual non-conforming items. The samples selected for test in Table II shall contain the samples selected from Table I and samples selected for test in Table III shall contain those samples selected for test outlined in Table IV.

4.7.1.1 Disposition of rejected lots.- Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lots shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

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TABLE I - CLASSIFICATION OF DEFECTS

Use Inspection Level L8, and Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: NONE DEFINED

MAJOR: AQL 1.5%	Req't Par.	QA. Par.	Test Method
101. Operating temperature	3.5.1	4.8.1.1	Standard Measuring Equipment (SME)
102. Vibration	3.5.2	4.8.1.2	SME
103. Torque	3.9.1	4.8.5	SME
104. Force	3.9.2	4.8.6	SME

MINOR: NONE DEFINED

TABLE II - CLASSIFICATION OF DEFECTS

Use Inspection Level II, and Sampling Plan Table IV -A of Standard MIL-STD-105

CRITICAL: NONE DEFINED

MAJOR: AQL 1.5%	Req't Par.	QA.Par.	Test Method
101. Deflection movement	3.6.1	4.8.2.1	Device, Special Testing (DST)
102. Elevation movement	3.6.2	4.8.2.2	DST
103. Backlash, boresight movement	3.7	4.8.3	DST
104. Boresight locks	3.8	4.8.4	DST
105. Alinement error (boresight locks)	3.8	4.8.4.1	DST
106. Elevation lock	3.8.1	4.8.4.2	DST
107. Torque	3.9.1	4.8.5	Standard Measuring Equipment (SME)
108. Force	3.9.2	4.8.6	SME
109. Identification	3.10	4.8.7	Visual

MINOR: NONE DEFINED

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TABLE III - CLASSIFICATION OF DEFECTS

Use Inspection Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: NONE DEFINED

MAJOR: AQL 2.5%	Req't Par.	QA. Par.	Test Method
101. Container size	5.2	--	Visual
102. Preservation missing, inadequate (Level A)	5.3.1.3	--	MIL-P-116
103. Barrier material and padding missing, inadequate (Level A,C)	5.3.1.4.1.4, 5.3.1.4.2, 5.3.1.4.1.6	--	Visual
104. Plywood base not conforming to specification (Level A,C)	5.3.1.4.1.3, 5.3.1.4.2, 5.3.2.2		Visual
105. Proper container	5.4.1.1		Visual
106. Mount not immobilized on plywood base (Level A,C)	5.3.1.4.1.5, 5.3.1.4.2		Visual
107. Plywood base (with packaged mount) not immobilized in shipping container (Level A,B,C)	5.3.1.4.1.3, 5.3.1.4.2, 5.3.2.2		Visual
108. Markings illegible, incorrect, missing, not conforming (Level A,B)	5.5		Visual
109. Improper cleaning (Level A,C)	5.3.1.1, 5.3.2.1		MIL-P-116
110. Improper drying (Level A,C)	5.3.1.2, 5.3.2.1		MIL-P-116
111. Packaging of instrument light	5.3.1.4.1.2		Visual
112. Shipping container immobilized (Level A,B)	5.4.1.2, 5.4.2, 5.4.3		Visual

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TABLE III (Cont'd)

MINOR: AQL 4.0%	Req't Par.	Test Method
201. Coated hardware not properly taped, improperly applied (Level A,C)	5.3.1.4.1.1, 5.3.1.4.2, 5.3.2.2	Visual
202. Strapping improperly applied, missing (Level A,B,C)	5.4.4	Visual
203. Workmanship (Level A,B,C)	5.7	Visual

TABLE IV - CLASSIFICATION OF DEFECTS

Use Inspection Level II, and Sampling Plan Table V of Standard MIL-STD-105

CRITICAL: NONE DEFINED

MAJOR: AQL 1.5%	Req't Par.	QA. Par.	Test Method
1. Heat seal (Level A)	5.3.1.4.1.7	4.9.12	Visual
2. Quick leak (Level A)	5.3.1.4.1.7	4.9.12	Visual

MINOR: NONE DEFINED

4.8 Test methods and procedures.

4.8.1 Environmental.

4.8.1.1 Operating temperatures.- The mount shall be placed in a test chamber capable of maintaining temperatures specified in 3.5.1. The test shall be conducted in accordance with Procedure I of Specification MIL-F-13926 and shall meet the requirements of 3.5.1 and 3.9.1.

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4.8.1.2 Vibration.- The mount shall be secured on a vibrating machine capable of vibrating the mount at the amplitude, duration and frequency specified in 3.5.2. At the conclusion of this test, the mount shall be visually inspected for damage or physical failure and shall meet the requirements specified in 3.6, 3.7, and 3.8 when tested by the method outlined herein.

4.8.2 Alinement.- Alinement of the mount shall be determined when the mount is positioned as specified in 3.6 by means of a special testing device the accuracy of which must be within the limits described in 4.6.2.1.

4.8.2.1 Deflection movement.- Actuation of the deflection boresight knob shall move the line of sight specified in 3.6 to the minimum distance specified in 3.6.1 as measured by means of a test target.

4.8.2.2 Elevation movement.- Actuation of the elevation boresight knob shall move the line of sight specified in 3.6 to the minimum distance specified in 3.6.2 as measured by means of a test target.

4.8.3 Backlash boresight movement.- This test shall be performed with the aid of an auxiliary sighting device. The line of sight of the sighting device when properly installed in the mount shall be directed toward a plumbed target containing a vertical and horizontal crossline. Rotate the line of sight in a continuous clockwise direction by means of the deflection boresight knob of the mount until the vertical reticle line is in coincidence with the vertical target line. Record reading on micrometer scale. Rotate the line of sight past the vertical target line (clockwise) a minimum of one full turn of the deflection knob, then by a continuous counterclockwise rotation of the deflection knob, duplicate the recorded micrometer setting; at this point the vertical reticle line shall be in coincidence with the vertical target line. Repeat these operations for elevation knob using the horizontal target line. The backlash shall not exceed the limits specified in 3.7 within the limits specified in 3.7. All errors shall be read on the target.

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4.8.4 Boresight locks.- The locking ability of the boresight knobs shall be determined after the auxiliary telescopes optical axis has been alined to a selected target point provided, and after the locking torque specified in 3.8 has been applied. The actual test is performed after applying the specified torque of 3.8. Any displacement between the points of coincidence shall be made by means of the target mil block. This displacement shall be within the limits of 3.8.

4.8.4.1 Alinement error resulting from actuation of boresight locks.- This test shall be conducted concurrent with the aforementioned test. The deflection and elevation alinement of the mount shall meet the requirements of 3.8 when the boresight locks are actuated with the torque specified. Any movement in the line of sight shall be means of mil block target.

4.8.4.2 Elevation locking lever.- When the elevation locking lever is released the elevation boresight knob shall move the line of sight as specified in 3.8.1.

4.8.5 Torque, boresight knobs.- This test shall be conducted at room temperature (plus 60 degrees to plus 90 degrees Fahrenheit) using a torque wrench capable of measuring the required torque. The torque required to rotate the boresight knobs shall meet the requirements of 3.9.1, and when subjected to the temperature of 3.5.1, shall meet the requirements of 3.9.1.

4.8.6 Force, disengagement of boresight nob from detent.- This test shall be conducted by means of an adapter and testing device specified in 4.6.2.4. The direction in which the force (pull) is applied shall be as specified in 3.9.2 and shall meet the requirements for standard ambient temperature specified in 3.9.2 and also shall meet the requirements for extreme temperatures of 3.9.2.

4.9 Packaging.

4.9.1 Rough handling test.- The pilot pack only shall be subjected to the rough handling test in accordance with the requirements of Specification MIL-P-116, applicable only to Level A.

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4.9.2 Packaging test.- Each package in the applicable sample shall be subjected to the pertinent packaging test of Specification MIL-P-116.

4.9.3 Determination of cleanliness test.- The contractor shall subject the applicable surfaces of one or more units to the "Determination of Cleanliness" test in accordance with Specification MIL-P-116.

4.9.4 Preproduction pack.- The preproduction pack shall be inspected by the contractor and witnessed by the Government inspector to determine conformance with the requirements of this specification for the packaging and packing level specified.

4.9.4.1 Preproduction pack failure.- Should the preproduction pack fail to pass the acceptance tests, production of those components causing rejection will cease until the defective characteristics have been corrected in the process used to produce the preproduction pack. When corrections in the process have been made, a new preproduction pack shall be completely inspected by the contractor and witnessed by the Government inspector.

5. PREPARATION FOR DELIVERY

5.1 Materials and methods.- The materials, methods and procedures used in the packaging of items shall be in accordance with this specification. Materials and methods not covered by a specification shall be suited to the intended use and subject to the approval of the contracting officer.

5.1.1 Materials.- The following shall apply wherever packaging materials are referred to in this specification by name only:

Reference	Full identification
Filament tape	Type II, 3/4 inch wide of Specification PPP-T-97
Greaseproofed barrier	Type I or II, grade A, Class 2 of Specification MIL-B-121
Gummed tape	Class 2 of Specification UU-T-111
Paperboard cushioning material	Type I, class I of Specification PPP-P-291
Pressure-sensitive tape	Type I or II of Specification UU-T-106

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5.2 Size of containers.- Bags and exterior containers shall be of a minimum size that will adequately contain the items.

5.3 PRESERVATION AND PACKAGING.

5.3.1 Level A.

5.3.1.1 Cleaning.- Cleaning Process C-1 of Specification MIL-P-116 shall be followed to clean all surfaces of the mount, instrument light and loose items.

5.3.1.2 Drying.- Immediately after cleaning, the surfaces shall be dried throughly in accordance with any applicable procedure specified in Specification MIL-P-116 that will not damage the items.

5.3.1.3 Preservation application.- All unprotected metal surfaces including the mounting hardware and locating pins, shall be coated with preservative conforming to Specification MIL-L-3150 (P-7 of Specification MIL-P-116). Preservative shall be applied on the unprotected metal surfaces of the mount by the brushing procedure specified in Specification MIL-P-116. Preservative shall be applied on the pins and hardware by any applicable procedure specified in Specification MIL-P-116. Excess preservative shall be removed.

5.3.1.4 Unit packaging.

5.3.1.4.1 Mount with instrument light.- The mount with the instrument light shall be packaged as specified herein, which is in conformance with Method IA-16 of Specification MIL-P-116.

5.3.1.4.1.1 Covering preservative coated surfaces and securing mounting hardware to the mount.- The preservative coated surfaces of the mount shall be covered or wrapped with greaseproofed barrier material which shall be secured in place with pressure-sensitive tape. The coated pins shall be wrapped with greaseproofed barrier material. The coated hardware and the wrapped pins shall be placed together and wrapped with grease-proofed barrier material, which shall be secured in place with pressure-sensitive tape. The wrapped hardware shall be secured to the mount with filament tape.

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5.3.1.4.1.2 Instrument Light, M50.- The instrument light shall be wrapped with paperboard cushioning material, which shall be secured in place with gummed tape. The wrapped instrument light shall be secured to the mount with filament tape.

5.3.1.4.1.3 Plywood base.- The base, which the mount is secured to, shall be fabricated from 3/4 inch thick plywood conforming to type I, II or III, class 2 of Specification NN-P-515. The length and width of the base shall be such that after the mount is secured to the base inside a sealed floating water-vaporproof bag (specified in 5.3.1.4.1.7) and the base is immobilized in the shipping container, a clearance of at least two inches shall exist between the maximum projection on the sides, ends and top of the bagged mount and the interior surfaces of the sides, ends and top panels of the shipping container. A 3/4 inch thick by 1-1/2 inches wide wood cleat shall be nailed to the bottom of the plywood base at each end of the base. The length of the cleats shall be approximately equal to the width of the base. Bolt holes shall be provided in the base for the maximum diameter step bolts that can be inserted in the mounting bolt holes in the telescope mount for bolting the mount to the base.

5.3.1.4.1.4 Water-vaporproof barrier and gaskets.- Water-vaporproof barrier material, conforming to class I of Specification MIL-B-131, shall be used in fabricating the floating bag specified in 5.3.1.4.1.7. Appropriate size bolt holes shall be made in the barrier material to permit the bolts to pass through for securing the mount to the base. Gaskets with appropriate size bolt holes for the bolts to pass through shall be cemented to each side of the water-vaporproof barrier in such a manner that the holes in the gaskets and the holes in the water-vaporproof barrier coincide. The gaskets shall be fabricated from material not less than 3/32 inch thick and conforming to class II, grade 60 of Specification MIL-R-6855. The adhesive used to cement the gaskets in place shall conform to type II of Specification MIL-A-5092. The length and width of the gaskets shall be approximately equal to the length and width of the plywood pad specified in 5.3.1.4.1.5. The aforementioned adhesive shall be applied on the surface of the gasket which will contact the base prior to placing the water-vaporproof barrier on the base. The water-vaporproof barrier shall then be positioned on the base in such a manner that the bolt holes in the base, the barrier and gaskets coincide.

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5.3.1.4.1.5 Securing mount to plywood base.- A plywood pad, fabricated from 1/2 inch thick plywood conforming to the characteristics of the plywood specified in 5.3.1.4.1.3 shall be positioned on the gasket which is located on the water-vaporproof barrier material covering the base. The size of the pad shall be approximately 4-1/2 inches by 6 inches and contain appropriate size bolt holes for the mounting bolts to pass through. The top surface of the pad shall be covered with greaseproofed barrier material which contains appropriate size bolt holes. The mount shall be positioned on the greaseproofed barrier material covered pad and secured in place with zinc or cadmium coated step bolts, flat washers, lockwashers and nuts. The bolts shall be the maximum diameter that will fit in the bolt holes in the mount. Unthinned paint shall be applied on the exposed threads above the nuts.

5.3.1.4.1.6 Covering projections, sharp edges and corners.- All projections, sharp edges and corners, that would puncture the water-vaporproof barrier when it is formed into a bag, shall be covered with paperbpard cushioning material. The paperboard cushioning material shall be secured in place with pressure-sensitive tape.

5.3.1.4.1.7 Water-vaporproof bag.- The water-vaporproof barrier material shall be heat sealed in such a manner as to form a bag that will pass the heat seal and quick leak tests of Specification MIL-P-116. Entrapped air shall be exhausted prior to final closure.

5.3.1.4.2 Unit packaging the mount without instrument light.- The mount shall be packaged as specified in 5.3.1.4.1.1 to 5.3.1.4.1.7 inclusive, except that the instrument light is omitted.

5.3.2 Level C.

5.3.2.1 Cleaning and drying shall be as specified in 5.3.1.1 and 5.3.1.2.

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5.3.2.2 Unit packaging.- The mount, with and without an instrument light, shall be packaged as specified in 5.3.1.4.1.1 to 5.3.1.4.2 inclusive, except as follows, which is in conformance with Method III of Specification MIL-P-116:

(a) Since preservative is not required, paper material, conforming to type II of Specification MIL-P-130, shall be used in lieu of greaseproofed barrier material specified under Level A unit packaging.
(b) The water-vaporproof barrier, gaskets and adhesive, specified in 5.3.1.4.1.4 shall be omitted.
(c) The protective coating on the step bolts, flat washers, lock-washers and nuts, specified in 5.3.1.4.1.5, shall be omitted.
(d) The covering of projections, sharp corners or edges specified in 5.3.1.4.1.6 shall be omitted.
(e) Since the water-vaporproof barrier is omitted, 5.3.1.4.1.7 is not applicable.

5.4 Packing.

5.4.1 Level A.

5.4.1.1 Shipping container.- One unit packaged mount, with or without an instrument light, shall be immobilized in a container (specified in 5.4.1.2) conforming to one of the following specifications:

PPP-B-601 - Style A or B (oversea type)-type II or III, class 2 plywood
PPP-B-621 - Class 2 (oversea), style 2 or 4

5.4.1.2 Immobilizing in shipping container.- Each plywood base containing a packaged mount, with or without an instrument light, shall fit snugly in the shipping container and shall be immobilized with a wood hold-down cleat in each corner of the container. The hold-down cleats shall be nailed to the end panels of the container. Each hold-down cleat shall be at least 3/4 inch thick by three inches wide and the length shall be such that the end of the cleat is flush with the top edge of the container. Cement coated nails shall be used. The size of the nails and nailing pattern shall conform to the box specification.

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5.4.2 Level B.- One unit packaged mount, with or without an instrument light, shall be immobilized in a shipping container as specified in 5.4.1.2. The shipping container shall conform to one of the following specifications:

PPP-B-601 - Any style under domestic type
PPP-B-621 - Class 1 (domestic), style 2 or 4

5.4.3 Level C.- One unit packaged mount, with or without an instrument light, shall be immobilized in a shipping container as specified in 5.4.1.2. The shipping container shall conform to one of the following specifications:

PPP-B-601 - Any style under domestic type
PPP-B-621 - Class 1 (domestic), any style except 1, 5 and 6.

5.4.4 Strapping.- Strapping shall be applied on each shipping container as specified in the applicable container specification.

5.5 Marking.

5.5.1 Interior packages.- Each floating bag shall be marked in accordance with Standard MIL-STD-129. The serial number shall be added in the identification data on each Level A unit package.

5.5.2 Shipping containers.- Each shipping container shall be marked in accordance with Standard MIL-STD-129. The serial number shall be added in the identification data on each Level A or B shipping container and omitted on the Level C container.

5.5.3 Stock number.- The stock number to be added in the identification data shall be obtained from the contracting officer.

5.6 Preproduction pack.- Prior to quantity packaging production, a preproduction pack, consisting of a complete and packed shipping container, shall be inspected to determine conformance with the applicable requirements of this specification, a rough handling test us required.

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5.7 Workmanship.- All operations involved in accomplishing corrosion prevention, preservation, packaging and packing by the procedures specified herein, shall be in accordance with the highest grades of practice associated with this type of work.

6. NOTES

6.1 Intended use.- The Mount Telescope, M107 is used to support the Articulated Telescope, M102 in the 120 MM Gun Tank, M103A1. The mount provides for boresight adjustments in elevation and deflection.

6.2 Order data.- Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Selection of an applicable level of preservation packaging and packing
(c) Applicable stock number
(d) Provisions for production model including date of delivery (see 3.1)

6.3 Size and weight of packed shipping container.- The following inside dimensions, overall cubage and gross weight are approximate for one unit packaged mount packed in a nailed wood box for Level A:

Inside dimensions - 30. inches long by 24. inches wide by 12-1/2 inches deep
Cubage - 6.8 cubic feet
Gross weight - 115. pounds

6.4 Design information.- Information regarding specific design not covered or referenced by this specification may be obtained from the procuring activity upon specific request of the contractor.

6.5 Definitions.

6.5.1 Standard measuring equipment(SME).- Standard measuring equipment includes the common hand type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment.

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6.5.2 Special testing device(DST).- Special testing devices are defined as special inspection fixture instruments and modified standard measuring equipment.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any person or corporation, or conveying any rights or permission to manufacture, use or sell any invention that may in any way be related thereto.

Custodian:

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